

Nathan Cook

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TECHNICAL SUMMARY

Computer Engineer specializing in applied machine learning, computer vision, and embedded systems. Experience in engineering robust hardware-software subsystems and deploying real-time AI architectures.

EDUCATION

Queen's University | Smith Engineering
Bachelor of Applied Science, Computer Engineering

Kingston, ON
May 2026

Relevant Coursework: Computer Networks, Computer Vision w/ Deep Learning, Machine Learning & Deep Learning, Artificial Intelligence, Safety Critical Software Engineering, Software Performance Analysis, Digital Systems

WORK EXPERIENCE

City of Ottawa – Traffic Operations

Ottawa, ON

Co-op Student

May 2025 – August 2025

- Interpreted traffic data and modelled new intersection timing splits through SimTraffic.
- Travelled to on-site locations for data collection and construction verification.
- Structured the department's traffic studies in collaboration with technologists, engineers, and summer students.
- Performed structured engineering-related measurements and timing changes for improved traffic flow.

City of Ottawa – Traffic Operations

Ottawa, ON

Traffic Data Analysis Assistant

May 2024 – August 2024

- Gathered on-site data for vehicle, cyclist and pedestrian movements to facilitate signal timing changes.
- Updated engineering clearance drawings through Bentley and their respective pedestrian timings.
- Documented current intersection conditions including signal heads, cross-walk audibles, and signal timings.

PROJECT EXPERIENCE

Smart Car Security System

September 2025 – April 2026

- Trained a custom YOLOv8s model to detect and classify people and several dangerous objects.
- Detected and tracked threats using motion sensors and an AI model for object tracking through pan-tilt adjustments.
- Delivered live footage and threat logs through a tailored mobile Android application using Azure Blob storage.
- Collaborated in a 3-person team to design and integrate AI, hardware, and software subsystems into a final product.

Brake Light Safety Controller

September 2025 – December 2025

- Designed an Arduino-based brake light controller using FreeRTOS, Simulink, and CAN bus signal processing.
- Established ASIL classifications via HARA and Fault Tree Analysis, ensuring 50ms hardware fault detection.
- Mapped technical safety requirements into Simulink logic blocks to govern unique braking scenarios.
- Achieved 100% MC/DC in Simulink and validated error correction through robust fault injection testing.

Small Scale Autonomous Vehicle

January 2025 – April 2025

- Configured an embedded Linux environment on a Raspberry Pi, installing the OS and necessary libraries.
- Implemented autonomous lane-following and collision avoidance using ultrasonic and grayscale sensors.
- Developed a YOLOv8n model for custom street signage recognition for edge computing on a Raspberry Pi.

SKILLS

Programming Languages: Python, C, C++, Java, HTML, CSS, PHP, SQL, Verilog, R

Frameworks & Libraries: OpenCV, PyTorch, TensorFlow, NumPy, FreeRTOS, XGBoost, Scikit-learn, Pandas

Software Applications: GitHub, SolidWorks, MATLAB, Simulink, Linux, LTspice, MySQL, Google Colab

Languages: English (*Native*), French (*Proficient*)